

Language as Cognitive Exhaust

What Language Reveals About Thought Compression and Artificial Intelligence

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Abstract

Language is a communicable residue of cognition rather than a complete record of the process that produced it. This paper develops cognitive exhaust as a model for understanding the distance between lived conditions, internal representation, and linguistic expression. Large language models learn primarily from the third layer and infer the others from traces. As generated language becomes input to later human and machine cognition, this distance creates a recursive pathway through which semantic drift can enter culture while language remains fluent.

Core Claim

Language preserves traces of cognition, but it cannot carry the full perceptual, emotional, causal, and contextual structure from which expression emerged.

Language as Residue

A sentence is the endpoint of many processes that do not appear in the sentence. Perception selects features of an environment. Memory supplies prior structure. Emotion changes salience. The mind compresses these materials into a form that can be expressed to another person.

Cognitive exhaust names the residue left by that process. The term does not imply that language is worthless or accidental. Exhaust can reveal the operation that produced it. Language contains evidence of thought, but it is not a complete copy of thought.

Three Layers of Representation

The process begins with lived or perceived reality, but what reaches language has already been shaped along the way. Experience is filtered through attention, memory, emotion, and prior concepts before any of it becomes a communicable account. Language then compresses part of that internal representation into symbols that someone else has to interpret.

Meaning can shift at each step. Perception selects from what is available, cognition organizes what was selected, and language reduces that structure again. The reader or listener then has to rebuild an

internal model from the trace that remains. Communication works when enough of the original structure survives that sequence.

Human Cognition as Recursive Compression

Human cognition works by compressing experience into forms that can be reused. Memories become categories, and those categories begin shaping what we notice later. Explanations can harden into expectations, which in turn affect what receives attention. Over time, more of thought is built from earlier representations rather than from direct encounter.

That is what makes abstraction and learning possible, but it also creates a route for drift. If an early representation drops an important distinction, later thought may inherit the simplified version without access to what was lost. Language can then stabilize that simplification and make it available for other people to reuse.

What Language Models Learn

Large language models are trained primarily on linguistic expression. They encounter the visible residue of cognition rather than the embodied and situational processes that produced it. Patterns in text carry substantial information about those hidden processes, allowing models to reconstruct plausible concepts, intentions, and world structure.

The reconstruction remains indirect. Multiple experiences can produce similar language, and similar experiences can be expressed in different ways. When models generate language, they infer missing structure from statistical relationships among traces. Fluency shows that the linguistic pattern is well formed. It does not establish contact with the experience, intention, or condition that originally gave the pattern meaning.

The Recursive Cultural Loop

The influence now moves in both directions. Human experience is compressed into cognition and language. Language becomes training data. Models generate new language from that residue. People read the generated language and use it to organize later thought, expression, and decision making. Those expressions return to the information environment.

The resulting loop can preserve and extend knowledge. It can also amplify inherited simplifications. Generated reconstructions become increasingly difficult to distinguish from accounts grounded in direct experience. The culture begins reasoning from language that was produced from prior language while the original experiential reference grows more distant.

Semantic Fidelity in Co-Cognition

As human and machine cognition become coupled through external memory and generated language, semantic fidelity cannot be treated as a property of isolated outputs. It becomes a property

of the loop. The relevant question is whether the combined system retains ways to restore contact with experience, evidence, and consequence.

This requires provenance, access to primary material, preserved uncertainty, and deliberate return to nonlinguistic conditions. Without those constraints, coherence can circulate between humans and models as mutual confirmation even when both are operating on the same compressed residue.

Limits of the Model

Cognitive exhaust is a conceptual model, not a claim that thought exists in a pure form that language merely contaminates. Thought is already shaped by language, culture, and prior representation. Nor does the model imply that unexpressed experience is automatically more accurate. Its purpose is to mark the losses and reconstructions that occur between experience, cognition, and expression.

The model becomes useful when a system treats linguistic fluency as evidence that it has recovered the process behind the language. The trace may support inference. It does not remove the distance between the trace and its source.

Conclusion

Language gives artificial intelligence access to an enormous record of human cognition, but the record is compressed and incomplete. As generated language becomes part of future cognition, the distinction between experience and its linguistic residue becomes harder to recover. Semantic fidelity depends on keeping that distinction visible.